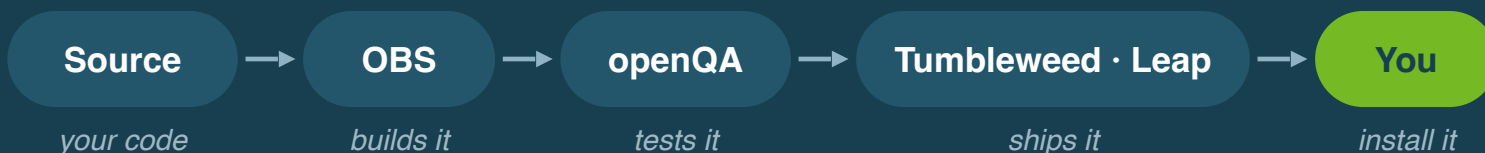


A BEGINNER-FRIENDLY OVERVIEW

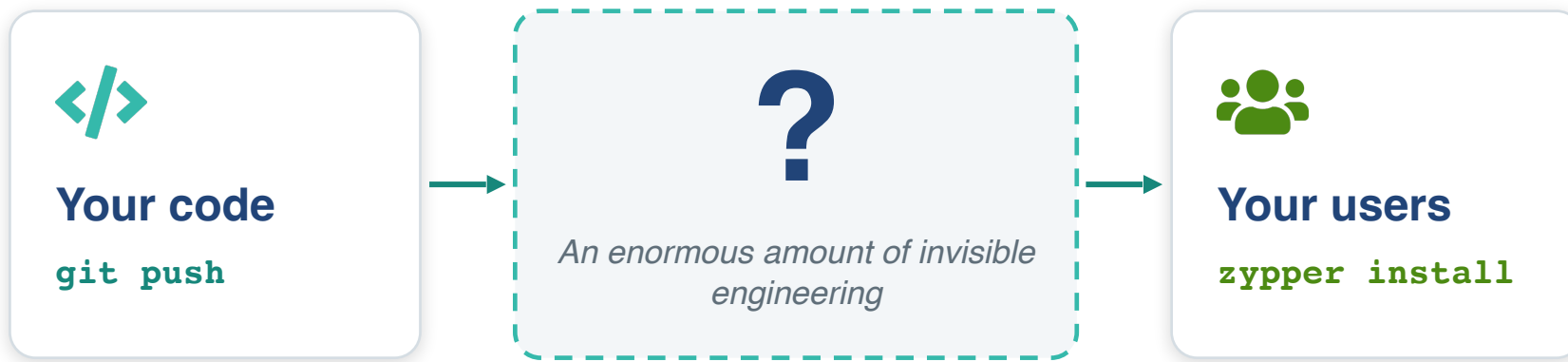
How openSUSE Works

From Source Code to Millions of Users



DIVYANSHU PRAKASH | RITHULL RAJ KALLAT

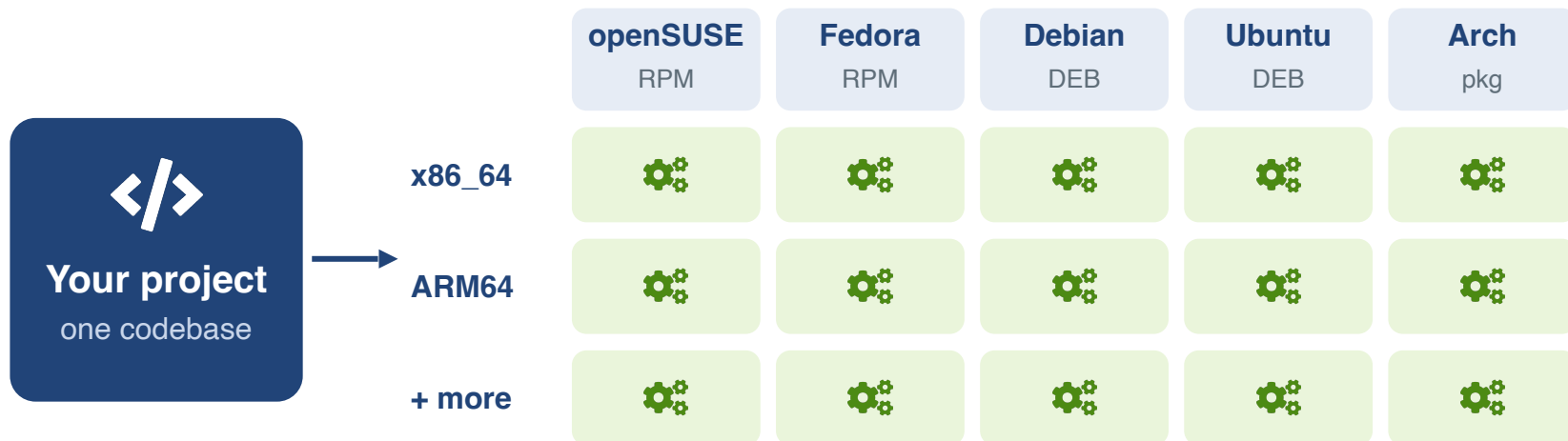
You Built It. Now What?



THE QUESTION FOR THIS TALK

What actually happens between `git push` and a user installing your software?

One Project. An Explosion of Targets.



5 distros × 3 architectures = **15 builds** for a single version.

Now multiply by dependency versions and distro releases.

Software distribution is a build problem.

“Works on My Machine.”



Your machine

Your compiler, cache and habits.



Your users' machines

A different distro. A different world.

..and every one of these can differ:

compiler versions

library versions

CPU architectures

dependency versions

distro releases

*What if your package assumes **libB v1** but the target ships **v2**?*

Humans aren't meant to repeat boring work perfectly, forever.

02 · THE IDEA

What If We Could Describe the Build?

Define it once. Build it every time, everywhere.

Meet the Open Build Service

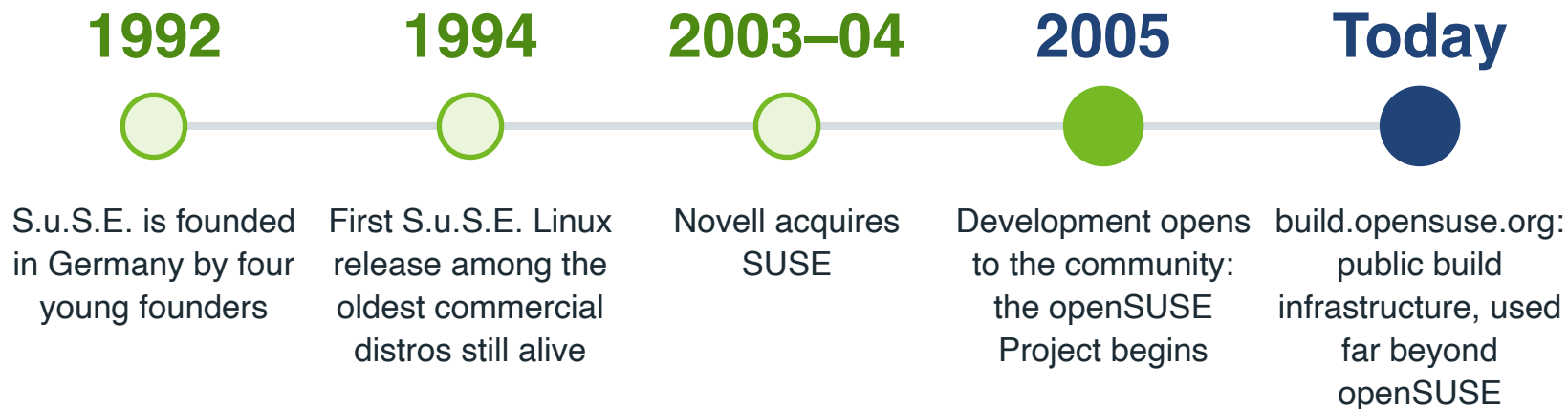
OBS is free software that builds packages from source automatically and consistently for many distributions and architectures, then publishes them as repositories anyone can install from.



build.opensuse.org: week to 31 Jul 2026 · 21 distributions / 6 architectures: openbuildservice.org/about

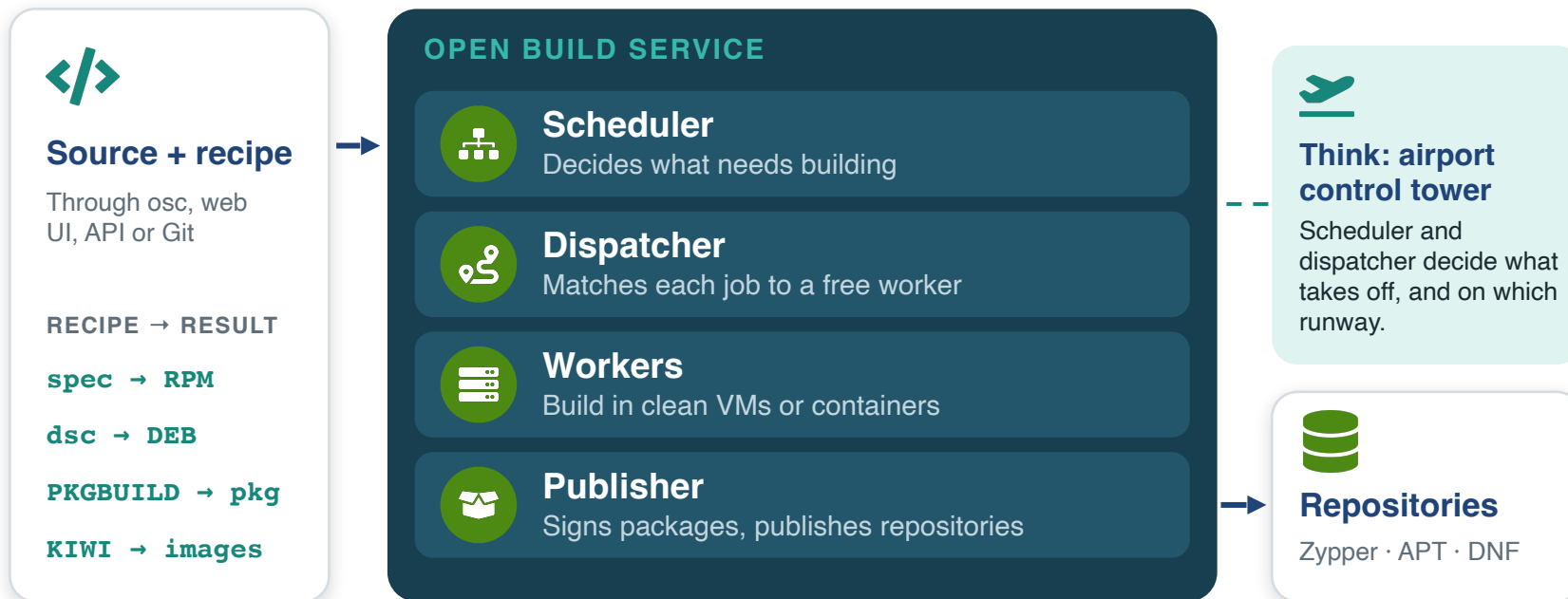
OBS turns “I built it on my laptop” into a repeatable, auditable pipeline.

Where OBS Comes From



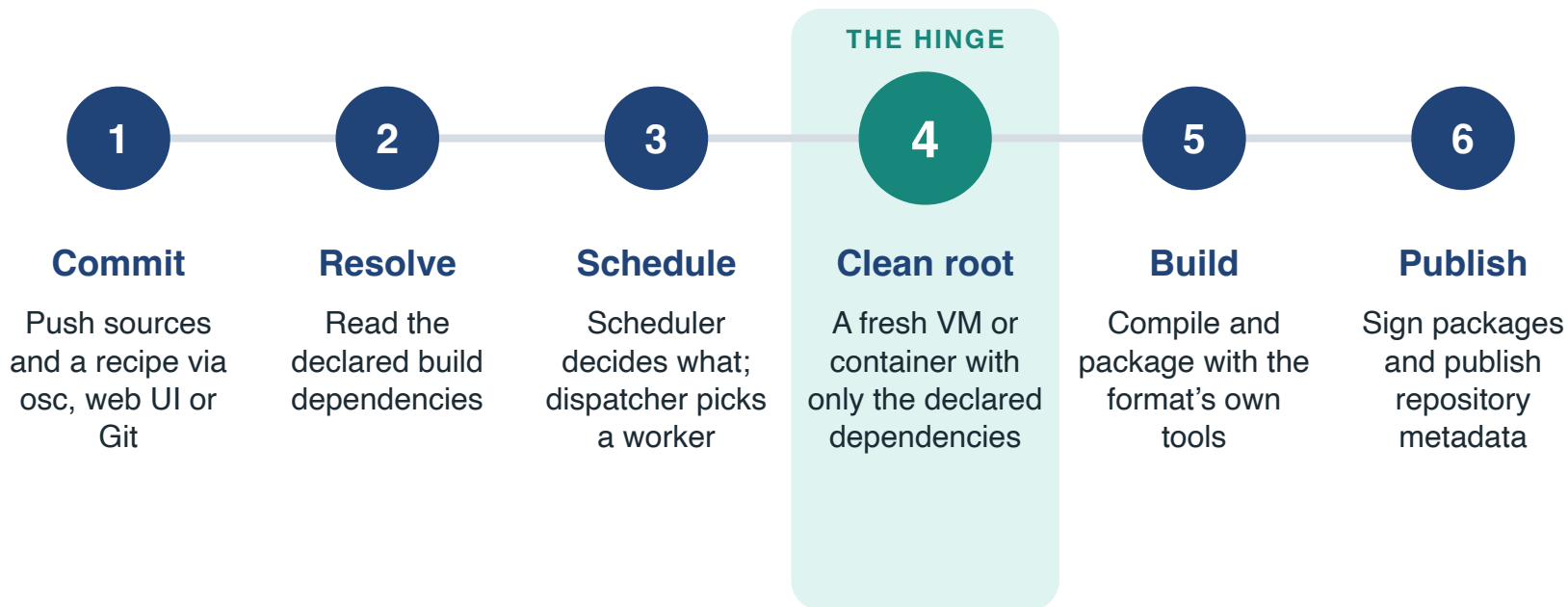
An open project needs open infrastructure.

OBS, in One Picture



One source description. Every target, reproduced automatically.

The Build Pipeline, End to End



Step 4 is the hinge: everything after depends on a clean environment.

Why Clean Build Environments Matter

YOUR LAPTOP

"It works because I have..."

- ✗ a random library
- ✗ an old dependency
- ✗ a custom config
- ✗ an environment variable
- ✗ a package I forgot about

OBS CLEAN BUILD

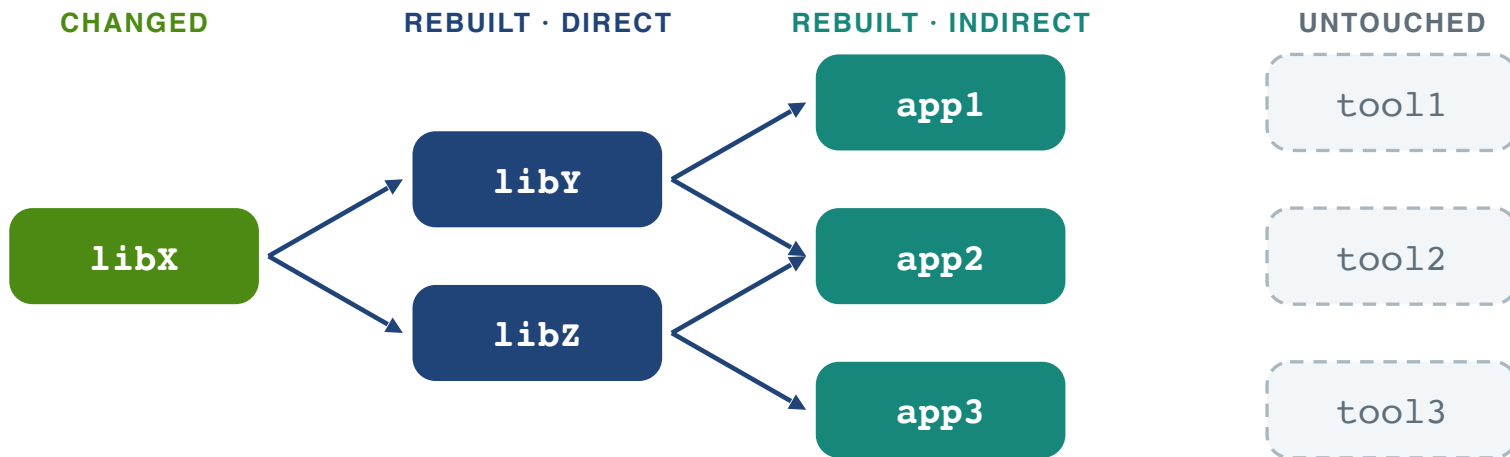
"Only declared dependencies."

- ✓ A fresh environment for every build
- ✓ Only the listed dependencies installed
- ✓ Nothing carried over between builds
- ✓ **Result: the build is honest**

Declared in the recipe: BuildRequires (spec), Build-Depends (debian/control)

Hidden dependencies are bugs waiting to happen.

A Change Ripples Through the Graph



By default OBS rebuilds all dependents (“transitive”):
a few extra builds buy a provably consistent state.

Change → dependency graph → rebuild what depends on it.

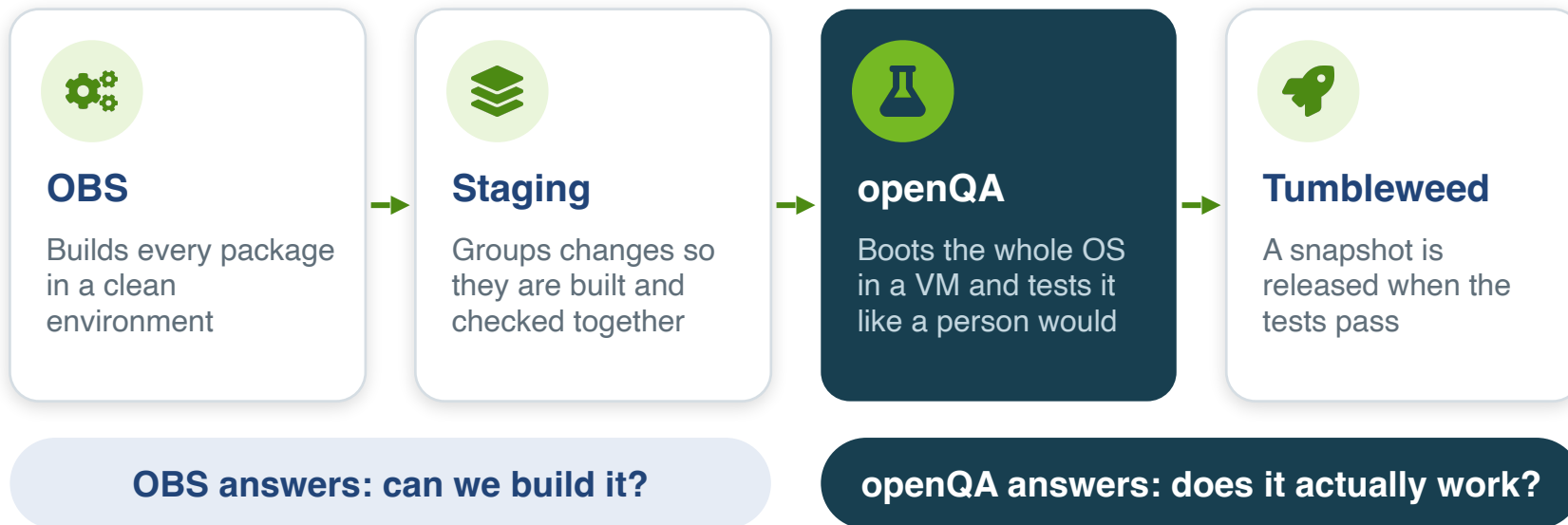
04 · ZOOM OUT

A green build means the build succeeded. Nothing more.

Build success is not the same as software correctness.



Who Tests the Result? Meet openQA.



Tumbleweed is the rolling release; Leap is the stable, regular release.

A green build proves the packaging worked. openQA checks the system works.

Isn't This Just CI?

	OBS	GitHub Actions · GitLab CI · Jenkins
FOCUS	Packages	Workflows
OUTPUT	Repositories for many distros and architectures	Test results, artifacts, deployments
LIVES IN	The package ecosystem	The GitHub / GitLab ecosystem

Also in this space:

Debian build

Fedora Koji and Copr

Launchpad PPAs

Complementary tools — OBS can even be triggered from GitHub or GitLab.

Open Infrastructure Builds Trust



Inspect

Anyone can read how packages are built.



Rebuild

Anyone can run the same build
OBS is free software (GPL).



Improve

Found a problem? Fix it once
everyone benefits.

Can someone else rebuild it?

YES → confidence

NO → mystery

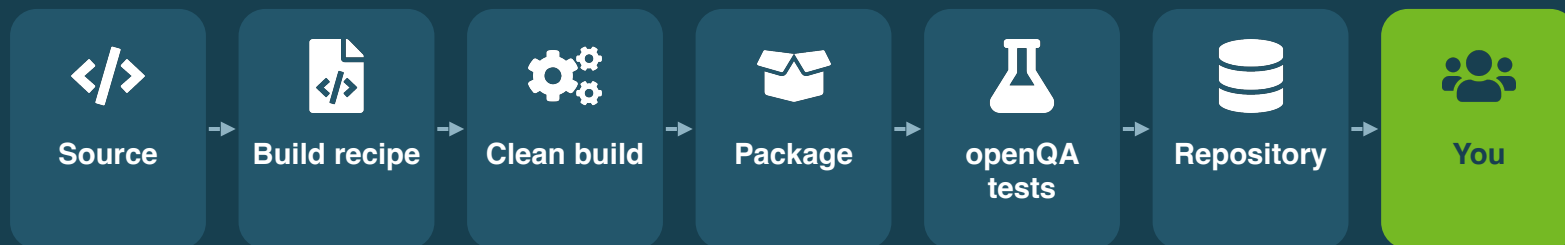
Repeatable builds mean easier debugging, maintenance and trust.

The Software You Install Has a Story

```
$ zypper install foo
```

what you see

what's underneath



That's what happens between `git push` and `zypper install`.

Distribution is engineering.

05 · YOUR TURN

“I’m just a student.”
Perfect.

Contribution doesn’t start with git clone and it doesn’t have to end
with git push.



Find a Problem. Help Solve It.

~~What can I code?~~ → **What problem can I solve?**

"This documentation is confusing."



Documentation

"This error message isn't clear."



Bug report or fix

"This package doesn't build."



Packaging

"This test is missing."



Testing

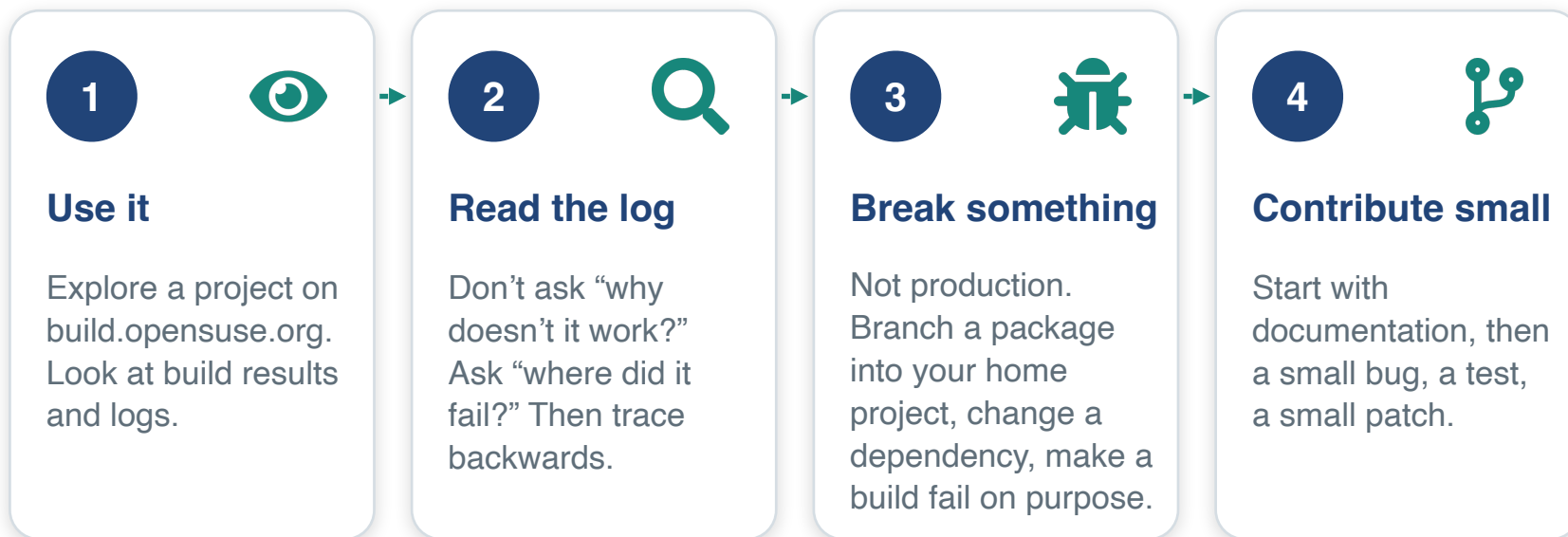
"This workflow is unnecessarily difficult."



Tools and community help

Pick one small problem, not the whole menu.

Four Steps In



Failed builds are tutorials.

Thank You

Questions?

Use it. Break it. Read the log. Contribute.



build.opensuse.org

Try it sign up and explore



openbuildservice.org

Documentation and news



github.com/openSUSE/open-build-service

Code and issues



openSUSE[®]
Asia Summit 2026

Divyanshu Prakash · Rithull Raj Kallat